ClinicNotes — draft consult notes from GP consultations transcripts with grounding

ELEC5623 Group Business Proposal · Track A

Group 21

Track A AI agent product. Group 21: Noah Williams 530502117, Zara Ahmed 530498620, Kevin Lin 530510342, Olivia Brown 530495876, Mateo Garcia 530507219. Tutor Dr K. Morrison approved 26/8/2026. Vision: GP's get a draft note that they can trust because every line points back to what was actually said.

1. Executive summary

General practitioners spend a huge amount of their working day doing documentation, research shows its close to half of a clinicians time is spent on the computer and documentation is one of the biggest causes of burnout, and after every consultation the GP has to write up the consult note which usually follows a SOAP structure (subjective, objective, assessment, plan) and this often happens at the end of the day from memory which leads to notes being incomplete or having errors, so what we propose is ClinicNotes which takes the transcript of a consultation (from an existing speech to text system, we dont do the speech recognition ourselves) and produces a draft SOAP note where every statement in the draft is linked to the transcript segment it came from, and statements that cant be linked are removed or flagged, and then the GP reviews and edits the draft and signs it off, so the GP stays in control. Ambient scribe products are already being adopted in Australian general practice so the idea is not new but our contribution is the verification layer and the measurement of how often drafts contain unsupported content, which is the thing clinicians and regulators worry about most.

2. Introduction and background

Consult notes are the legal record of a consultation and they are used for continuity of care, referrals, billing and medico-legal purposes. A note needs to record what the patient reported (subjective), what was examined or measured (objective), the GPs assessment and the plan including any prescriptions, referrals and follow up. Currently GPs type notes during or after the consultation, some use dictation and more recently some practices use AI scribes which listen to the consult and write a note automatically. There are several of these products in the Australian market such as Heidi Health and Lyrebird Health and internationally Nuance DAX. They are popular because they save time but there is concern that they sometimes include things that were not said (hallucinations) or miss important things, and the GP may not notice because the note reads fluently. Our understanding is that professional bodies have advised that the GP remains responsible for the content of the note and must check it.

3. Problem statement and motivation

The specific problem is that AI generated consult notes can contain content not supported by the consultation and the GP has no efficient way to check each statement against what was said, so either they read the whole transcript again (which removes the time saving) or they trust the draft (which is risky). The people affected are GPs, patients (whose records may be wrong) and practices (who carry medico legal risk). It matters because errors in records can affect future care for example a wrong allergy or a dose. GenAI is appropriate because turning a conversational transcript into a structured clinical note requires understanding of medical language, paraphrase and the structure of a consultation, which a rule

based system cant do, but the same capability creates the risk of unsupported content, which is why we add verification.

4. Stakeholders and requirements

The primary stakeholders are GPs (users who review and sign notes) and practice managers (who choose software). Secondary stakeholders are patients, the practice's clinical governance lead and the IT provider. GPs need a draft that is correct, complete and quick to check, practice managers need something that fits the workflow and complies with privacy obligations, patients need their record to be accurate and their data protected.

Scope: text transcripts of GP consultations in English, SOAP note drafting, statement level linking and a review screen. Not in scope: speech recognition, prescribing, integration with clinical software like Best Practice or MedicalDirector, specialist consultations, anything diagnostic.

Functional requirements (owner in brackets, each has an acceptance criteria): FR1 accept a transcript with speaker labels and timestamps (A1) — 100% of test transcripts load with speakers preserved. FR2 segment the transcript into utterances with ids (A1) — every utterance has an id and timestamp. FR3 detect and mask names, dates of birth and Medicare numbers before any model call (A1) — 100% of planted identifiers in test transcripts masked. FR4 draft a SOAP note from the transcript (A2) — every test transcript produces a note with all four sections. FR5 link every statement in the draft to one or more utterance ids (A2) — 100% of displayed statements carry at least one link. FR6 put explicitly negative findings into the note when stated, eg "no chest pain" (A2) — negations recorded in ≥90% of cases in the test set. FR7 verify each statement against its linked utterances with a separate entailment check (A3) — statements judged unsupported are flagged, flag precision ≥0.85. FR8 remove or flag statements with no valid link (A3) — no unlinked statement is shown without a flag. FR9 extract medications, doses and allergies into a structured list and check them against the transcript (A3) — 100% of medication names in the list appear in the transcript. FR10 show the draft and transcript side by side (A4) — clicking a statement scrolls to the linked utterances. FR11 let the GP edit, accept or delete any statement (A4) — edits saved with time and the original AI text kept. FR12 show unsupported and low confidence statements first (A4) — flagged statements listed at the top. FR13 export the signed note as text in the practice template (A5) — export matches template headings. FR14 log model version, prompt version and verification results for each note (A5) — every note has a complete log record. FR15 provide an evaluation mode that runs the full pipeline on a labelled set (A5) — one command reproduces all metrics.

Non functional requirements: NFR1 latency, draft and verification ready in under 60 seconds for a 15 minute consult. NFR2 grounding, 0% of displayed statements without a link. NFR3 privacy, identifiers masked before any external call and no transcript stored outside the local machine. NFR4 reproducibility, temperature 0 and logged prompt versions so a run can be repeated. NFR5 usability, GPs can review a note in under 3 minutes on average in the study. NFR6 cost, under 20 cents per note.

Constraints: C1 only synthetic transcripts, no real patient data at all. C2 the prototype runs on a laptop and only the model endpoint is external. C3 the semester budget for model use is $120. C4 no integration with clinical software.

5. Proposed product and innovation

The product is a web app where the GP opens a consult, sees the draft note on the left and the transcript on the right, statements are colour coded (linked and verified, linked but weakly supported, unsupported) and clicking any statement jumps to the utterances. The GP edits and signs. The innovation compared to

existing scribes is the statement level linking plus independent verification, where existing products give you a fluent note but not the evidence trail, and our evaluation will specifically measure unsupported content which vendors dont publish in a comparable way.

6. Methodology and system design

The pipeline works like this. First the transcript is parsed into utterances with ids and speaker labels and identifiers are masked by a combination of regular expressions (Medicare numbers, dates) and a named entity model for names, masked tokens are kept consistent so the note still reads correctly and they are unmasked only at display time locally. Second the drafting step sends the masked transcript to the LLM with instructions to produce a JSON object with four sections each containing a list of statements and for each statement the list of utterance ids that support it, the prompt includes the SOAP definitions and two worked examples from synthetic consults. Third the linker validates that every utterance id exists, any statement with no existing id is marked unlinked. Fourth the verifier takes each statement and its linked utterances and asks a second model call, with a different prompt and no access to the rest of the draft, whether the utterances entail the statement, contradict it or are insufficient, the answer is one of three labels, statements labelled insufficient or contradicted are flagged. Fifth medications, doses and allergies are extracted into a structured list and each item is string matched against the transcript (after normalising brand and generic names using a small lookup table), anything not found is flagged in red. Sixth the review interface shows everything and stores the GPs edits separately from the AI draft so we can measure how much the GP had to change. The architecture is a Python FastAPI backend with the pipeline steps as separate modules, SQLite for storage, a React front end, and a hosted LLM through an OpenAI compatible API with the option of a local open weights model for the verifier if cost or privacy requires it. We chose to use a separate verification call rather than asking the drafting model to self check because self checking tends to agree with itself. Development is in two week iterations, iteration 1 parsing masking and drafting, iteration 2 linking and verification, iteration 3 review interface, iteration 4 evaluation and refinement.

6.1 More detail on the SOAP rules we give the model

The subjective section is what the patient reports in their own words so symptoms, duration, history of the presenting complaint, relevant past history, medications the patient says they take and allergies the patient reports, and it should be written in third person ("reports a dry cough for 5 days") and not include the GPs interpretation. The objective section is only things the GP measured or observed during the consult, like blood pressure readings that were said out loud, temperature, examination findings that the GP verbalised ("chest is clear", "no tenderness over the spine") and it must never include anything the GP did not say because in a transcript we cant see what the GP saw silently, this is an important limitation and the note will say "examination findings not verbalised" if there is nothing. The assessment is the GPs working diagnosis or impression as they stated it and the plan is investigations, prescriptions with dose and frequency, referrals, advice given, safety netting ("come back if fever above 38.5 or breathing gets worse") and follow up timing. We found when trying early prompts that the model likes to add plausible advice that was never given (eg "advised rest and fluids") which is exactly the kind of unsupported content we want to catch so the prompt says explicitly that every plan item must be something the GP said and the verifier checks it.

6.2 Worked example of the pipeline on one consult

To make the design concrete here is how one of our synthetic consults goes through it. The consult is a 9 minute appointment for a 34 year old with a sore throat and fever for three days, the transcript has 64 utterances. Masking replaces the patients name in 3 places and the date of birth in 1 place. The drafting

step produces 19 statements, 7 subjective, 3 objective, 2 assessment, 7 plan, and links each to between 1 and 4 utterances. The linker finds that all ids exist. The verifier labels 17 statements as entailed, 1 as insufficient (the draft said "no cough" but the patient actually said "not really much of a cough" which the verifier decided was not a clear negative) and 1 as contradicted (the draft said "amoxicillin 500mg three times daily for 10 days" but the GP said "for 5 days"). The medication check also flags the duration. In the review screen these two statements appear at the top in amber and red, the GP clicks each, sees the utterances, corrects the duration and rewords the cough statement, and signs. Total review time in our dry run was about 70 seconds. Without the flags the wrong duration would have been easy to miss because the rest of the note reads fine.

6.3 Why two calls and not one

We considered asking the drafting model to output a confidence for each statement but model self reported confidence is not well calibrated in our experience and in what we have read, also a model checking its own output tends to agree with itself, so the verifier is a separate call that sees only the statement and the cited utterances and nothing else from the draft, this makes the verification independent of the drafting context and also makes it cheap because each verification input is short. Statements are verified in batches of 10 to reduce the number of calls. If the verifier output does not parse we treat the statements in that batch as unverified and flag them, we never silently pass them.

7. Product and business analysis

The value is a note that saves the GP time like other scribes do but that can be checked quickly because the evidence is right there. Users are GPs, customers are practices or practice groups who pay per clinician per month for software. Alternatives are typing notes manually (accurate if done carefully but slow and often done late), dictation (faster but still the GP composes the note), and existing AI scribes (fastest and already adopted, but no statement level evidence trail and limited published data on unsupported content). We think a realistic path is not to compete with these products directly but to show that a verification layer measurably reduces unsupported content, which could be licensed or adopted as a feature. Costs are model calls, about two calls per statement batch, estimated under 20 cents per note, plus hosting. Adoption would need privacy assessment and clinical governance approval which is outside the semester but we designed for it (masking, local storage).

More on alternatives because this is important for whether the project is worth doing. Manual typing is the baseline everyone knows, GPs are fast typists and use templates and abbreviations so a note might take 2 to 4 minutes but often it is done later in the day and things are forgotten. Dictation tools convert speech to text but the GP still has to compose the note which is the slow part. AI scribes do the composing and that is why they are being adopted quickly, the GP we spoke to (a family friend of one of the team) said she likes hers but she has caught it putting in examination findings she never did and now she reads every note carefully which takes away some of the time saving. So the gap is not drafting itself, it is checking, and that is the part we focus on. We also thought about whether a practice would pay for a separate checking product and honestly we think it is more likely to be a feature that scribe vendors add, but for this project the point is to show it works and measure it.

8. Evaluation plan

We will create 40 synthetic consult transcripts (20 dev and 20 test) by writing scenarios across common presentations (respiratory infection, hypertension review, back pain, mental health check in, medication review) and having two team members role play them and transcribing them. For each transcript we will write a reference note and a list of key facts that must appear. We will also plant errors for testing the

verifier by creating versions of drafts with inserted unsupported statements (wrong dose, symptom not mentioned, wrong side of body). Baselines are B1 manual note writing by a team member from the transcript (timed), and B2 a direct LLM draft with the same model but no linking or verification. Metrics are key fact recall (share of key facts present in the draft, target ≥0.90), unsupported statement rate (share of displayed statements not supported by the transcript as judged by two annotators, target ≤2% for ClinicNotes vs B2 measured), planted error detection (share of planted unsupported statements flagged, target ≥0.90), medication accuracy (target 100%), review time per note (target under 3 minutes with at least 3 volunteer reviewers) and edit distance between draft and final note. We will report results per presentation type and show failure examples.

9. Risks ethics and responsible AI

The biggest risk is clinical harm from an incorrect note that is signed without checking, mitigation is the verification flags, the review ordering that puts flagged statements first and making very clear in the interface that the GP is responsible. Privacy risk is handled by using only synthetic data in the project and masking identifiers before model calls. There is a risk that the verifier itself is wrong (false reassurance) which we measure through planted errors and report honestly. Bias risk exists if transcripts of patients with accents or non standard English produce worse drafts, we can only partly test this with synthetic data and will state it as a limitation. Technical risk is that linking fails on long consults, fallback is to chunk the transcript by topic. Schedule risk is that role playing transcripts takes longer than expected, fallback is to write them directly as text.

10. Plan and team

Week 5 to 6 parsing masking and drafting (Noah and Zara), week 7 to 8 linking and verification (Kevin), week 9 to 10 review interface (Olivia), week 10 to 11 medication check and export (Zara and Mateo), week 11 evaluation runs (Mateo with everyone labelling), week 12 reviewer study and fixes, week 13 presentation. Responsibilities: A1 Noah Williams parsing and masking, A2 Zara Ahmed drafting and medication extraction, A3 Kevin Lin linking and verification, A4 Olivia Brown review interface, A5 Mateo Garcia export logging and evaluation. Dependencies are that drafting must work before verification can be tested, if we fall behind we drop export formatting first and then the local model option.